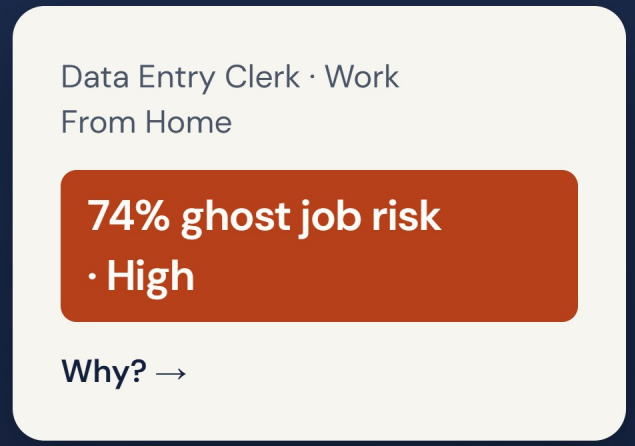
IBM Z DATATHON 2026 · THEME 2: AI SECURED

RealRole

A second opinion on every job listing, before you spend hours on a job that isn't real.



Rachel Chen · Olivia Moody · Michael Wang · Alan He

THE PROBLEM

Ghost jobs look exactly like real listings

Hours wasted

Tailored resumes and cover letters go to jobs that were never going to be filled.

Data exposed

Applicants hand over personal details to listings that lead nowhere.

No time to check

Students and new grads apply to hundreds of jobs and can’t vet each one.

[Optional: one verified statistic on ghost jobs, with its source]

THE IDEA

A risk badge right on the listing

Chrome extension on Greenhouse, Lever, Workday and LinkedIn

Lookup site: paste a job link or the listing's text

"Why?" shows the reasons; IBM Granite sums them up in one sentence

A probability with reasons, **never a verdict**



THE DATA

Real, labeled ghost job data is hard to find

17,880

listings in EMSCAD, the best open dataset

No public, current dataset labels real vs. ghost job listings.

866

labeled ghost jobs, only 4.8%

LinkedIn and Indeed ban scraping; commercial data costs enterprise prices.

2012–14

when they were posted

We cleaned it: 281 duplicates removed, text repaired, salary and location parsed.

Source: EMSCAD (Kaggle, CC0), University of the Aegean · ml/AUDIT.md

WHAT WE CAUGHT

A random split would have doubled our score



No company has both real and ghost job listings. A random split put 80% of test listings in companies the model had already seen, so it learned *who* posted, not *what* a ghost job looks like.

PR-AUC of a model trained on company-description text only · ml/AUDIT.md

THE MODEL

Predictive AI scores, generative AI explains

Clean
Text, TF-IDF and listing details; missing-value flags

→

LightGBM
Beat a logistic baseline; calibrated, so 70% means about 70%

→

TreeSHAP
Exact reasons per field, quoting words from the listing

→

IBM Granite
One plain sentence from the reasons; never changes the score

Tuned honestly: 20 configurations on 5 company-grouped folds; test set scored once, at the end

Live-page aware: fields a page can't show are left out, not counted as "missing"

Safe wording: a check blocks any accusation and falls back to a template

Uses both IBM labs: hybrid multi-model predictive AI, and predictive + generative AI

RESULTS ON COMPANIES THE MODEL NEVER SAW

Catches most ghost jobs, rarely bothers real ones

0.739
PR-AUC, about 15× better than chance

68%
of ghost jobs caught

2 in 3
flags are real ghost jobs

1.8%
of genuine listings wrongly flagged

Live check: **200 real listings** from public companies scored through the extension. Median risk 0.6%, **none flagged**.

Held-out test: 2,514 listings, 122 ghost jobs, scored once · 95% CI for PR-AUC 0.57–0.89 · docs/model-card.md

IBM LINUXONE

Private by design, on IBM Z

Your browser

Extension or lookup site sends only the listing's public text

→

RealRole API on LinuxONE

Own Podman container, the only published port; LightGBM and TreeSHAP in about 150 ms

→

IBM watsonx.ai

Granite 3 8B writes the one-sentence summary

Nothing about you is sent: no browsing history, no profile, no identity

Nothing is stored: the API never logs requests or pasted links

Theme 2: innovation without exposing the job seeker

[Add LinuxONE screenshots once deployed: uname -m → s390x, podman ps, curl /score]

WHAT'S NEXT

With LinkedIn data, RealRole could protect every job seeker on LinkedIn

Reported listings

Which postings users flagged and LinkedIn removed

Recruiter signals

Verification, account age and reposting history

Real outcomes

How applicants respond, and whether roles get filled

Fresh, labeled listings at scale, instead of 12-year-old data: a far more accurate and far more useful app.

HONEST LIMITS

What we know it gets wrong, and what's next

Limits today

Training data is from 2012–2014

Polished listings with a logo can slip through

Genuine remote, part-time and no-logo listings are flagged more (8–11% vs. 1.8%)

Next steps

Partner with LinkedIn for fresh, labeled data

Full LinuxONE deployment with Granite 8B on watsonx

Publish the extension on the Chrome Web Store

RealRole

Explainable, calibrated and private: a second opinion before you spend your time on a job that isn't real.

Rachel Chen
Data, split, API, extension, lookup site

Olivia Moody
LightGBM, tuning, calibration, threshold

Michael Wang
Features, baseline, LinuxONE platform

Alan He
Evaluation, metrics, integration

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